Arithmetic Expressions

CHAPTER 02

Learning Outcomes...

- Simple Arithmetic Expressions
- Comparing Expressions
- Arithmetic Expression in Real Life
- Reading and Evaluating Complex Expressions
- Terms in Expressions
- Swapping and Grouping
- Removing Brackets in Expressions
- Tinker the Terms

Introduction

An arithmetic expression is defined as mathematical phrase that involves numbers and mathematical operators (such as +, –, ×, ÷) and sometimes brackets, arranged in a meaningful way to show a mathematical calculation.

For example ; 5 + 4, 12 – 4 × 2, (8 + 4) ÷ 6

On the basis of numbers of operations involved, the arithmetic expressions are divided into two categories:

(i) Simple Expressions (ii) Complex Expressions

Simple Arithmetic Expressions

It involves only one operation, either addition, subtraction, multiplication or division and no brackets.

For example ; 2 + 5, 6 × 3, 12 – 5, 16 ÷ 4, etc.

Value of an Expression

Each arithmetic expression represents a value, which is the result, we get after performing the operations.

For example ; Value of the expression 24 – 2 is 22, which is obtained by subtracting 2 from 24.

The expression 24 – 2 can be read as 24 minus 2 or subtraction of 2 from 24, etc.

The equal sign (=) is used to show the connection between an arithmetic expression and its value.

For example ; 24 – 2 = 22

Note that different expressions can have same value. For example; 8 + 2 = 10, 12 – 2 = 10, 5 × 2 = 10, and 20 ÷ 2 = 10. So, there 4 expressions have same value or we can say that 10 can be expressed in 4 different ways by using two numbers and any of the operator (+, –, ×, ÷).

Comparing Expressions

Comparison of the expression can be done is two ways:

(i) By comparing the values directly: When we compare expressions, we check if they are equal (=), greater (>) or less (<) in value and place the sign (=, > or <) accordingly between expressions.

For example; $7 + 5 \square 14 - 1$

As $7 + 5 = 12$ and $14 - 1 = 13$

Since 12 is less than 13, so we have $7 + 5 \boxed{<} 14 - 1$

(ii) By logical reasoning: The comparison of two simple expressions can also be done through logical reasoning by analysing the differences between the numbers involved. This method saves time and improves mental math skills.

For example; Consider the expressions: $245 + 289 \square 246 + 285$

Here, 245 is 1 less than 246.

and 289 is 4 more than 285.

So, the first number is 1 less but the second number is 4 more, resulting in a net gain of 3.

Hence, $245 + 289 \boxed{>} 246 + 285$

ILLUSTRATIONS

1 **Fill in the blanks to make the expressions equal.**

(i) $12 + 6 =$ ____ $+ 9$

(ii) $54 - 12 = 16 +$ ____

(iii) $55 \div$ ____ $= 5 \times 1$

Soln.: (i) $12 + 6 = 18$

So, ____ $+ 9 = 18 \Rightarrow$ ____ $= 18 - 9 = 9$

$\therefore\ 12 + 6 = \underline{9} + 9$

(ii) $54 - 12 = 42$

So, $16 +$ ____ $= 42$

$\Rightarrow$ ____ $= 42 - 16 \Rightarrow$ ____ $= 26$

$\therefore\ 54 - 12 = 16 + \underline{26}$

(iii) $5 \times 1 = 5$

So, $55 \div$ ____ $= 5$

$\Rightarrow$ ____ $= 55 \div 5 = 11$

$\therefore\ 55 \div \underline{11} = 5 \times 1$

2 **Find the values of the given expressions and compare them by using sign >, < or =.**

(i) $15 + 3 \square 17 + 2$

(ii) $5 \times 5 \square 30 \div 6$

(iii) $25 - 5 \square 12 + 8$

Soln.: (i) $15 + 3 = 18$ and $17 + 2 = 19$

Since, $18 < 19$

$\Rightarrow 15 + 3 \boxed{<} 17 + 2$

(ii) $5 \times 5 = 25$ and $30 \div 6 = 5$

Since, $25 > 5$

$\Rightarrow 5 \times 5 \boxed{>} 30 \div 6$

(iii) $25 - 5 = 20$ and $12 + 8 = 20$

$\therefore\ 25 - 5 \boxed{=} 12 + 8$

3 Compare the following expressions without complicated calculations. Explain your thinking in each case.

(i) $654 + 239 \square 653 + 242$

(ii) $412 - 168 \square 411 - 165$

(iii) $532 - 219 \square 531 - 218$

Soln.: (i) 654 is 1 more than 653, but 239 is 3 less than 242. So overall, the second expression is greater by 2.

Thus, $654 + 239 \boxed{<} 653 + 242$.

(ii) 412 is 1 more than 411, but 168 is 3 more than 165. So, the first expression starts with more but loses more—overall, the second expression has more left.

Thus, $412 - 168 \boxed{<} 411 - 165$.

(iii) 532 is 1 more than 531, but 219 is also 1 more than 218. So both increased by the same amount and therefore no overall change.

Thus, $532 - 219 \boxed{=} 531 - 218$

Arithmetic Expression in Real Life

We use arithmetic expressions all the time. When we add, subtract, multiply or divide quantity or numbers in daily life situations, we are actually using arithmetic expressions.

For example ;

(i) A cinema has 20 rows of seats, each row has 15 seats. Then the expression representing total number of seats in cinema hall is 20×15, *i.e.*, the product of 20 and 15.

Now, $20 \times 15 = 300$. So, there is 300 seats in the cinema hall.

(ii) Riya buys 25 notebooks, she gave 10 notebooks to her brother. Then the expression representing the notebook left with Riya is $25 - 10$, *i.e.*, 25 minus 10.

Now, $25 - 10 = 15$. So, Riya left with 15 notebooks.

Reading and Evaluating Complex Expressions

Sometimes, when an arithmetic expression is not given with a clear context, it may seem like there is more than one way to solve it. This can lead to different answer from different people, even though the numbers and operators look the same.

For example ; the expression $8 \times 5 + 2$ can be solved in two ways :

- $8 \times 5 + 2 = 40 + 2 = 2$
- $8 \times 5 + 2 = 8 \times 7 = 56$

It is obvious that only one of the above is correct but which one? And how do we decide?

In such situations, we need some rules to guide us the correct order of solving expressions, so that we get the same answer every time.

While finding a value of the expression; we do operations in the following order :

(i) First solve the expression inside brackets.

(ii) Next, do division and multiplication (from left to right).

(iii) Finally, do addition and subtraction (from left to right).

For example ; Rohan went to an amusement park with his friends. He bought 3 ride tickets, each costing ₹ 50 and 1 snack combo worth ₹ 100. He also get a discount of ₹ 20 on the total bill.

So, the expression which represent the money spend by Rohan in total is equal to

$3 \times 50 + 100 - 20$

Let us find the value of this arithmetic expression.

(i) Multiplication : $3 \times 50 = 150$

(ii) Addition : $150 + 100 = 250$

(iii) Subtraction : $250 - 20 = 230$

So, total amount spend by Rohan is ₹ 230.

ILLUSTRATIONS

4 Aman saves ₹ 50 per week for 8 weeks. He gave ₹ 150 to his sister. How much money does he left?

Soln.: Amount saved by Aman in 8 weeks $= ₹\ (50 \times 8)$

Amount given to his sister = ₹ 150

Expression representing the amount, Aman left with $= ₹\ (50 \times 8 - 150) = ₹\ (400 - 150)$ $= ₹\ 250$.

5 Solve the given expressions:

(i) $25 + (5 \times 6)$

(ii) $55 - (4 + 3) \times 6$

Soln.: (i) $25 + (5 \times 6) = 25 + 30 = 55$

(ii) $55 - (4 + 3) \times 6 = 55 - 7 \times 6 = 55 - 42 = 13$

6 Create a real life story or situation that matches each of the following arithmetic expressions. Then, calculate the value of each expression:

(i) $7 \times 15 - 10$ (ii) $30 - 5 \times 6 + 12$

Soln.: (i) Aryan bought 7 notebooks each costing ₹15 in total bill he got a discount of ₹ 10, then the total amount paid by Aryan is $7 \times 15 - 10 = 105 - 10 = ₹95$

(ii) Kabir was playing a board game with his friends. He started with 30 points, then he lost 6 points in each of the 5 rounds due to wrong answers. Later, he won a bonus of 12 points for completing a challenge.

At the end, the points that Kabir had

$= 30 - (6 \times 5) + 12 = 30 - 30 + 12 = 0 + 12 = 12$

Additive Inverse of a Number

The additive inverse of a number is the number that, when added to the original, results in zero. In other words, it's the opposite of the number in sign.

Every number has an additive inverse.

If we take a number 'a', its additive inverse is $-a$, as $a + (-a) = 0$.

Thus, the additive inverse of a positive number is negative and the additive inverse of a negative number is positive.

For example ;

(i) Additive inverse of 12 is –12, as $12 + (-12) = 0$

(ii) Additive inverse of –7 is 7, as $(-7) + 7 = 0$

Terms in Expressions

When we write an expression, it is usually made up of numbers and operations. To understand an expression properly, it's important to break it into smaller parts. These smaller parts are called terms.

Terms are the individual parts of an expression that are separated by addition (+) sign.

For example ; In the expression 25 + 8, these are two terms *i.e.*, 25 and 8.

Subtracting a number is the same as adding its inverse.

Consider the expression; 90 – 30. So, instead of writing 90 – 30, we can write it as 90 + (–30).

Here, the expression 90 – 30 has two terms: 90 and –30.

Note : The expressions like 6×5 or $12 \div 6$ are considered as single terms, because there is no '+' or '–' sign separating the parts of the expression. Multiplication or division alone does not break an expression into multiple terms.

ILLUSTRATIONS

7 Find the inverse of the following numbers.

(i) 36 (ii) –25 (iii) 36×21 (iv) $-(10 \times 5)$

Soln.: (i) The inverse of 36 is –36.

(ii) The inverse of –25 is 25.

(iii) The inverse of 36×21 is -36×21.

(iv) The inverse of $-(10 \times 5)$ is 10×5.

8 Find the number of terms in the given expressions and also identify each term.

(i) $4 + 7 \times 8$ (ii) $15 + 6 \div 3 - 2$

Soln.: (i) $4 + 7 \times 8 = 4 + (7 \times 8)$, there are 2 terms: 4 and (7×8)

(ii) $15 + 6 \div 3 - 2 = 15 + (6 \div 3) + (-2)$, there are 3 terms: 15, $(6 \div 3)$, –2

Swapping and Grouping

Swapping Property or Commutative Property of Addition

The swapping property or commutative property states that :

Changing the order of the terms does not effect the result, and the sum will remain the same. *i.e.*, for any two terms a and b; $a + b = b + a$

For example ; the expression with terms 12 and (–3), $12 + (-3) = (-3) + 12 = 9$

Grouping Property or Associative Property of Addition

The grouping property or associative property states that grouping the terms in any way (using brackets) does not effect the result, and the sum will still be the same. For any three terms a, b and c, $(a + b) + c = a + (b + c)$

For example ; the expression with terms 3, 6 and 4,

$(3 + 6) + 4 = 9 + 4 = 13 \Rightarrow 3 + (6 + 4) = 3 + 10 = 13$

Thus, $(3 + 6) + 4 = 3 + (6 + 4)$

COMPETITION WINDOW

- Commutativity of terms does not hold with respect to subtraction.
  For example ; $9 - 4 \neq 4 - 9$
- Associativity of terms does not hold with respect to subtraction.
  For example ; $(5 - 3) - 1 = 2 - 1 = 1 \Rightarrow 5 - (3 - 1) = 5 - 2 = 3$
  So, $(5 - 3) - 1 \neq 5 - (3 - 1)$

ILLUSTRATIONS

9 "Insert suitable numbers and operations sign in the blanks to make both sides of the expression equal."

(i) 15 + ___ = 9 + 15

(ii) (8 + 4) + (– 2) = ___ + (4 ___ (–2))

Soln.: (i) Using commutative property of addition for the terms 9 and 15,

$15 + 9 = 9 + 15$

Thus, the missing number is 9.

(ii) Using associative property of addition for terms 8, 4 and (–2),

$(8 + 4) + (-2) = 8 + (4 + (-2))$

Thus, the missing number is 8 and sign will be +

10 In each group of expressions given below, find the ones that are equal in value – without actually calculating them.

(i) $156 - 564$, $564 - 156$, $561 + 156$, $-564 + 156$

(ii) $(121 - 67) + 45$, $121 + ((-67) + 45)$, $121 + (45 + 67)$

Soln.: (i) $156 - 564 = 156 + (-564) = (-564) + 156$ (by using commutative property of addition)

(ii) $(121 - 67) + 45 = (121 + (-67)) + 45$
$= 121 + ((-67) + 45)$ (by associative property of addition)

Removing Brackets in Expressions

When we write arithmetic expressions, we often use brackets to group numbers or terms. These brackets help us decide the order in which operations should be done.

But sometimes, we need to remove the brackets to simplify or rewrite an expression. While doing this, we must pay attention to the sign (plus or minus) just before the bracket.

The effect of removing brackets depends on the sign just before the bracket.

When a plus (+) sign appear just before a bracket, the sign inside the bracket remains the same.

For example ; $8 + (3 + 2) = 8 + 3 + 2$

When a minus (–) sign appears just before a bracket, the signs inside the bracket change.

For example ; $8 - (3 + 2) = 8 - 3 - 2$

Because subtracting a group is the same as subtracting each number in the group individually.

When a multiplication (×) sign appears just before a bracket, multiply that number with each terms inside the bracket keeping the sign inside the bracket same.

For example : (i) $3 \times (4 + 2) = 3 \times 4 + 3 \times 2$

(ii) $3 \times (4 - 2) = 3 \times 4 - 3 \times 2$

(iii) $(5 + 7) \times 8 = 5 \times 8 + 7 \times 8$

(iv) $(7 - 4) \times 5 = 7 \times 5 - 4 \times 5$

It means that multiplying a sum (or difference) by a number gives the same result as multiplying each terms separately and then adding (or subtracting) those products. Thus, this property is also known as distribution property of multiplication over addition or subtraction.

Thus, for any numbers a, b and c;

$a \times (b + c) = a \times b + a \times c$ or $a \times (b - c) = a \times b - a \times c$

The distributive property of addition over multiplication for any numbers a, b and c :

$(a \times c) + (b \times c) = (a + b) \times c$

COMPETITION WINDOW

- The distributive property of addition or subtraction over multiplication

  For any numbers a. b and c :

  $(a \times c) + (b \times c) = (a + b) \times c$ or $(a \times c) - (b \times c) = (a - b) \times c$

ILLUSTRATIONS

11 Sam has ₹ 500. He buys 3 books priced at ₹ 120 each and 3 bookmarks costing ₹ 25 each. Write an expression with and without brackets representing how much money sam has spent. Also, find the amount of money Sam has left.

Soln.: Amount spent by Sam on books

= ₹ 3×120

Amount spent by Sam on bookmarks

= ₹ 3×25

Expression representing total amount spent by Sam

$= 3 \times 120 + 3 \times 25 = 3 \times (120 + 25)$

Amount of money that Sam has left

$= 500 - 3 \times (120 + 25)$

$= 500 - 3 \times 145 = 500 - 435 = 65$

12 A boy had 50 chocolates. He gave 18 to his sister and 7 to his friend.

(i) Write an expression using brackets to find the number of chocolates left with boys.

(ii) Remove the brackets and then find the number of chocolates left with the boy.

Soln.: (i) Expression representing number of chocolates that boy left with $= 50 - (18 + 7)$

(ii) Removing bracket :

$50 - (18 + 7) = 50 - 18 - 7 = 32 - 7 = 25$

Tinker the Terms : Creating New Expressions from the Given Expression

Sometimes, we are given a simple arithmetic expression – and by adding, removing, or modifying its terms, we can create a new expression with a different meaning or result. This process of adjusting or building expressions is called tinker the terms. While tinkering a term, we can find the value of new expression by using the previous one.

Change a Number in the Addition or Subtraction

Suppose we have given the value of an expression, $a + b = c$, then what will be value of expression $(a + 1) + b$?

It will be simply $c + 1$ because $(a + 1) + b = a + b + 1 = c + 1$

Similarly, $(a - 1) + b = c - 1$ or $a + (b - 1) = c - 1$

For example ; $156 + (-42) = 114$ so $155 + (-42) = 114 - 1 = 113$

This will be helpful while dealing with bigger numbers.

Change a Number in the Multiplication

Sometimes, we already know the answer to a multiplication like 45×12. But if we are asked to find 55×12, then we do not need to start calculation again.

We can use the multiplication we already know and adjust it based on how the numbers have changed. If one number increases or decreases, then we can use addition or subtraction to make a new expression and solve it easily.

For example ; $45 \times 12 = 540$, then $55 \times 12 = (45 + 10) \times 12 = 45 \times 12 + 10 \times 12$

$= 540 + 120 = 660$

ILLUSTRATIONS

13 A box of 5 pencils costs ₹ 120. Find the price of 10 such boxes. If a shopkeeper wants to buy 12 such boxes, then find the cost of 12 such boxes using the "Tinker the term" method.

Soln.: Cost of one box = ₹ 120

Cost of 10 boxes = 10×120 = ₹ 1200

Cost of 12 boxes = $12 \times 120 = (10 + 2) \times 120$

$= 10 \times 120 + 2 \times 120$

= 1200 + 240 = ₹ 1440

So, cost of 12 boxes will be ₹ 1440.

14 A factory produces 62 toys per day. In 30 days, it made 62×30 *i.e.*, 1860 toys. Next month, the factory increased production to 65 toys per day. Find how many toys it will make in next 30 days using the "Tinker the term" method.

Soln.: A factory produces 62 toys per day. In one month (30 days), it produced,

$62 \times 30 = 1860$ toys

Next month, the production increases to 65 toys per day. We have to find : 65×30

Thus, $65 \times 30 = (62 + 3) \times 30 = 62 \times 30 + 3 \times 30$
$= 1860 + 90 = 1860 + 90 = 1950$

Hence, the factory will produce 1950 toys in next 30 days.

CONCEPT MAP

ARITHMETIC EXPRESSIONS

Expressions formed by numbers and mathematical operations like +, –, × and ÷.
Example : 13 + 2, 20 × 4 – 3

Value of an Expression

The number we get after evaluating the expression is the value of that expression.
We use '=' sign to show the value.
For example ; 20 × 4 – 3 = 80 – 3 = 77

Comparing Expression

Comparison of the expression can be done in 2 ways:
(i) by comparing the value directly
(ii) by analysing the difference between the number involved.

Evaluating Complex Expression

We do expressions in the following order
(i) First solve expression inside brackets
(ii) Next do division and Multiplication (from left to right)
(iii) Finally, do addition and subtraction (from left to right)

Use of Brackets

Brackets clarify which operation to perform first.
For example ; 30 + (5 × 4)
= 30 + 20 = 50

Removing Brackets

- When a plus (+) sign appears just before a bracket, the signs inside the bracket remains the same.
  For example ; 8 + (3 + 2) = 8 + 3 + 2
- When a plus (–) sign appears just before a bracket, the signs inside the bracket changes.
  For example ; 8 – (3 + 2) = 8 – 3 – 2
- When a multiplication (×) sign appears just before a bracket, multiply that number with each term inside the bracket keeping the sign inside the bracket same.
  For example ; 3 × (4 + 2) = 3 × 4 + 3 × 2

Terms in Expression

Individual parts separated by '+' sign.
For example ; 13 – 2 + 6
= 13 + (–2) + 6
∴ These are three terms *i.e.*, 13, –2, 6

Swapping and Grouping Terms

Commutative Property of Addition (swapping) : Changing order of terms does not changes the result.
For example ; 6 + (–4) = (–4) + 6 = 2
Associative Property of Addition (Grouping) : Grouping terms in any way (using brackets) does not change the result.
For example ; (–7 + 10) + (–11)
= –7 + (10 + (–11))

Tinker the Terms

- Change a number in addition or subtraction :
  For any two terms say a and b if $a + b = c$, then $(a + 1) + b = a + b + 1 = c + 1$ and $(a - 1) + b = c - 1$
- Change a number in multiplication : Use previously known value to find the value of expression by using distributive property.
  For example ; We know that 25 × 4 = 100, so
  26 × 4 = (25 + 1) × 4
  = 25 × 4 + 4 = 100 + 4 = 104

Solved Examples

1. For each of the situations given below, from the corresponding expression, identify its terms and find the value of the expression.

(i) Seema has ₹ 750. She spent ₹125 on groceries. How much money is she left?

(ii) Neha has ₹1000. She gave ₹250 to her brother and won ₹150 in a quiz. How much money she had now?

(iii) A movie ticket costs ₹90. Sanyam has purchased 5 tickets and one food combo of ₹ 150. How much money does Sanyam had spent?

Soln.: **(i)** Seema had ₹750 out of which she spent ₹125 on groceries

∴ Money left with Seema = ₹(750 – 125)

The expression is 750 – 125, which has 2 terms *i.e.*, 750 and (–125).

Now, 750 + (–125) = 625

∴ Seema left with ₹ 625.

(ii) Expression representing the given situation is ₹(1000 – 250 + 150) = ₹ (1000 + (–250) + 150)

It has three terms *i.e.*, 1000, (–250) and 150

Now, 1000 – 250 + 150 = 750 + 150 = 900

∴ Neha had ₹ 900 at the end.

(iii) Amount spend on tickets = ₹ 5 × 90

Amount spend on food combo = ₹ 150

Expression representing amount spent

= ₹(5 × 90 + 150)

It has 2 terms *i.e.*, (5 × 90) and 150

Now, (5 × 90) + 150 = 450 + 150 = ₹ 600

Thus, Sanyam has spend ₹ 600.

2. Find the values of the given arithmetic expressions by removing brackets and write them all in ascending order.

(i) 12 × 3 – (6 + 2)

(ii) (10 + 2) × 5

(iii) 15 – 3 – (6 × 2)

Soln.: **(i)** 12 × 3 – (6 + 2) = 36 – 6 – 2

= 30 – 2 = 28

(ii) (10 + 2) × 5 = 10 × 5 + 2 × 5 = 50 + 10 = 60

(iii) 15 – 3 – (6 × 2) = 12 – 12 = 0

Now, 0 < 28 < 60 so, (iii) < (i) < (ii)

3. Verify the commutative property for 9 + (–4) = (–4) + 9.

Soln.: 9 + (–4) = 9 – 4 = 5

(–4) + 9 = 5

∴ 9 + (–4) = (–4) + 9

So, commutative property holds.

4. Verify the associative property for (3 + 5) + 7 and 3 + (5 + 7).

Soln.: (3 + 5) + 7 = 8 + 7 = 15

and 3 + (5 + 7) = 3 + 12 = 15

∴ (3 + 5) + 7 = 3 + (5 + 7)

So, associative property holds.

5. A shelf has 4 rows with 18 books in each row. Find the number of books in the shelf. Another shelf has 4 more books per row and number of rows are same. Find the total number of books in the new shelf using "Tinker the Terms" method.

Soln.: Number of books in shelf

= 4 × 18 = 72 books

Number of books in new shelf = 4 × (18 + 4)

= 4 × 18 + 4 × 4

= 72 + 16 = 88 books

6. A library ordered 7 bundles of 12 books each and later, added 3 bundles of 12 books each. Write an expression to find the total number of books they ordered. Hence, using property to evaluate the value of the expression.

Soln.: Number of books in 7 bundles ordered by library = 7 × 12

Number of books order in 3 bundles = 3 × 12

Expression for total number of books ordered = (7 × 12) + (3 × 12).

Using distributive property of addition over multiplication, we get (7 × 12) + (3 × 12)

= (7 + 3) × 12 = 10 × 12 = 120

Hence, total number of books ordered are 120.

7. A volunteer raised ₹ 1500 in total donated by 6 persons equally. She later found 4 more sponsors donated the same amount. Write an expression for the total money raised and hence find the value of total amount.

Soln.: Money raised by 6 persons = ₹1500

Money raised by 1 person = ₹(1500 ÷ 6) = ₹250

Total money raised = 6 × 250 + 4 × 250

= (6 + 4) × 250 = 10 × 250 = ₹2500

8. Fill in the blanks using logical reasoning, without actual calculation, so that the expressions on both sides are equal.

(i) 312 + 8 = 308 + ____

(ii) 150 – 47 = 148 – ____

Soln.: (i) Since, 312 is 4 more than 308.

Therefore, to make the given expressions equal, the second number of RHS must be 4 more than second number of LHS.

Hence, 312 + 8 = 308 + 8 + 4 = 308 + 12.

(ii) Since, 150 is 2 more than 148.

Therefore to make the given expression equal, the second number of LHS must also be 2 more than second number of RHS.

∴ 150 – 47 = 148 – (47 – 2)

= 148 – 45

9. Kiran is not great with numbers, so whenever she wants to subtract 8, she forgets how much to take aways. Instead she remove 10 and add back 2 to balance it.

(i) Is this trick clever thinking?

(ii) How much she add back if she needs to subtract 9 and had removed 10?

Soln.: (i) She wants to subtract 8. Instead, she subtract 10 and then adds back 2.

Mathematically,

Number – 8 = (Number – 10) + 2

This is a valid arrangement so her trick is a clever thinking.

(ii) As 9 = 10 – 1

⇒ –9 = – (10 – 1) = –10 + 1

So, number – 9 = (number – 10) + 1

Thus, if she wants to subtract 9 and had removed 10, then she needs to add back 1.

10. Consider the given expression

(i) 68 × 14 – 3

(ii) 70 × 14 – 2 × 14 – 3

(iii) 68 × 10 + 68 × 4 – 3

Show that all the given expression are equal by using properties.

Soln.: 68 × 14 – 3 = (70 – 2) × 14 – 3

= 70 × 14 – 2 × 14 – 3

Also, 68 × 14 – 3 = 68 × (10 + 4) – 3

= 68 × 10 + 68 × 4 – 3

Thus, all the given three expressions are equal.

11. Given that 525 × 16 = 8400, use this result to find the value of 515 × 16.

Soln.: 515 × 16 = (525 – 10) × 16

= 525 × 16 – 10 × 16

= 8400 – 160 = 8240

12. Three friends use different subtraction tricks to subtract a same number.

Aarav : Subtracts 10, adds 6

Mehak : Subtracts 5, adds 1

Zoya : Subtracts 8, adds 4

What number are they trying to subtract?

Soln.: Aarav subtracts 10 and add 6 *i.e.*,

Number – 10 + 6 = Number – 4

Mehak subtracts 5 and add 1 *i.e.*,

Number – 5 + 1 = Number – 4

Zoya subtracts 8 and add 4 *i.e.*,

Number – 8 + 4 = Number – 4

∴ They all trying to subtract 4.

13. If 116 × 15 = 1740 and 116 × 5 = 580, then find 110 × 20 by using the given values.

Soln.: We have, 116 × 15 = 1740 and 116 × 5 = 580

Now, 110 × 20 = (116 – 6) × (15 + 5)

= (116 – 6) × 15 + (116 – 6) × 5

= (116 × 15) – (6 × 15) + (116 × 5) – (6 × 5)

= 1740 – 90 + 580 – 30

= 1650 + 580 – 30

= 2230 – 30 = 2200

14. Add brackets at appropriate places in the expressions, so that the expressions gives the value mentioned.

(i) 56 – 14 + 9 = 33

(ii) 25 – 16 + 2 × 8 = 56

Soln.: **(i)** If we solve it without adding brackets we have,

$56 - 14 + 9 = 42 + 9 = 51 \neq 33$

As $51 > 33$ so, we need to subtract more from 56. Let us try $56 - (14 + 9) = 56 - 14 - 9 = 42 - 9 = 33$, which indicate the given value.

(ii) Let us first solve it without adding brackets.

$25 - 16 + 2 \times 8 = 25 - 16 + 16 = 25 \neq 56$

Let us try $25 - (16 + 2) \times 8$

$= 25 - 18 \times 8 = -119 \neq 56$

Now, $(25 - (16 + 2)) \times 8$

$= (25 - 18) \times 8 = 7 \times 8 = 56$

15. In each case, insert the correct symbol (<, >, or =) in the box to show the relationship between them. Use your understanding of brackets, number operations, and mathematical properties to reason through the comparisons and do not calculate the exact values.

(i) $(6 - 2) \times 15 \ \square\ (2 - 6) \times 15$

(ii) $12 + 4 \times 17 \ \square\ (12 + 4) \times 17$

(iii) $(40 - 35) \times 9 \ \square\ (40 \times 9) - (35 \times 9)$

Soln.: **(i)** $(6 - 2) \times 15$ = Positive number × 15

And $(2 - 6) \times 15$ = Negative number × 15

Since, positive number > negative number

$\therefore \quad (6 - 2) \times 15 \boxed{>} (2 - 6) \times 15$

(ii) Since, $(12 + 4) \times 17 = 12 \times 17 + 4 \times 17$.

The expression $(12 + 4) \times 17$ means 17 times of both the numbers *i.e.*, 12 and 4. However, the expression $12 + 4 \times 17$ means 17 times of only the number 4 and 1 time of the number 12.

$\therefore \quad 12 + 4 \times 17 \boxed{<} (12 + 4) \times 17.$

(iii) $(40 - 35) \times 9 \boxed{=} 40 \times 9 - 35 \times 9$

[Using distributive property]

NCERT Section

Figure it Out *(Page 25)*

1. Fill in the blanks to make the expressions equal on both sides of the '=' sign:

(a) $13 + 4 =$ _____ $+ 6$

(b) $22 +$ _____ $= 6 \times 5$

(c) $8 \times$ _____ $= 64 \div 2$

(d) $34 -$ _____ $= 25$

2. Arrange the following expressions in ascending (increasing) order of their values.

(a) $67 - 19$ (b) $67 - 20$

(c) $35 + 25$ (d) 5×11

(e) $120 \div 3$

Figure it Out *(Page 34 & 35)*

1. Find the values of the following expressions by writing the terms in each case.

(a) $28 - 7 + 8$ (b) $39 - 2 \times 6 + 11$

(c) $40 - 10 + 10 + 10$ (d) $48 - 10 \times 2 + 16 \div 2$

(e) $6 \times 3 - 4 \times 8 \times 5$

2. Write a story / situation for each of the following expressions and find their values.

(a) $89 + 21 - 10$ (b) $5 \times 12 - 6$

(c) $4 \times 9 + 2 \times 6$

3. For each of the following situations, write the expression describing the situation, identify its terms, and find the value of the expression.

(a) Queen Alia gave 100 gold coins to Princess Elsa and 100 gold coins to Princess Anna last year. Princess Elsa used the coins to start a business and doubled her coins. Princess Anna bought jewellery and has only half of the coins left. Write an expression describing how many gold coins Princess Elsa and Princess Anna together have.

(b) A metro train ticket between two stations is ₹ 40 for an adult and ₹ 20 for a child. What is the total cost of the tickets:

(i) for four adults and three children?

(ii) for two groups having three adults each?

(c) Find the total height of the window by writing an expression describing the relationship among the measurements shown in the picture.

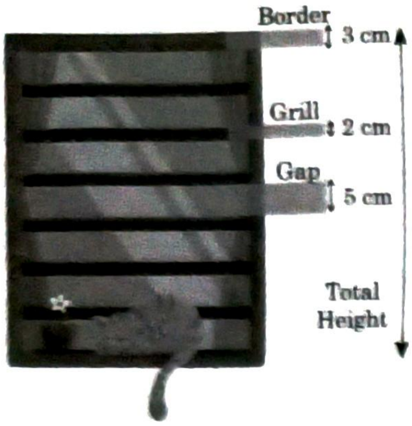


Figure it Out *(Page 37 & 38)*

1. Fill in the blanks with numbers, and boxes with operation signs such that the expressions on both sides are equal.

(a) 24 + (6 – 4) = 24 + 6 $\square$ _____

(b) 38 + (_____ $\square$ _____) = 38 + 9 – 4

(c) 24 – (6 + 4) = 24 $\square$ 6 – 4

(d) 24 – 6 – 4 = 24 – 6 $\square$ _____

(e) 27 – (8 + 3) = 27 _____ 8 _____ 3

(f) 27 – (_____ $\square$ _____) = 27 – 8 + 3

2. Remove the brackets and write the expression having the same value.

(a) 14 + (12 + 10)

(b) 14 – (12 + 10)

(c) 14 + (12 – 10)

(d) 14 – (12 – 10)

(e) –14 + 12 – 10

(f) 14 – (–12 – 10)

3. Find the values of the following expressions. For each pair, first try to guess whether they have the same value. When are the two expressions equal?

(a) (6 + 10) – 2 and 6 + (10 – 2)

(b) 16 – (8 – 3) and (16 – 8) – 3

(c) 27 – (18 + 4) and 27 + (–18 – 4)

4. In each of the sets of expressions below, identify those that have the same value. Do not evaluate them, but rather use your understanding of terms.

(a) 319 + 537, 319 – 537, -537 + 319, 537 – 319

(b) 87 + 46 – 109, 87 + 46 – 109, 87 + 46 – 109, 87 – 46 + 109, 87 – (46 + 109), (87 – 46) + 109

5. Add brackets at appropriate places in the expressions such that they lead to the values indicated.

(a) 34 – 9 + 12 = 13

(b) 56 – 14 – 8 = 34

(c) –22 – 12 + 10 + 22 = – 22

6. Using only reasoning of how terms change their values, fill the blanks to make the expressions on either side of the equality (=) equal.

(a) 423 + _____ = 419 + _____

(b) 207 – 68 = 210 – _____

7. Using the numbers 2, 3, and 5, and the operators '+' and '–', and brackets, as necessary, generate expressions to give as many different values as possible. For example, 2 – 3 + 5 = 4 and 3 – (5 – 2) = 0

8. Whenever Jasoda has to subtract 9 from a number, she subtracts 10 and adds 1 to it. For example, 36 – 9 = 26 + 1.

(a) Do you think she always gets the correct answer? Why?

(b) Can you think of other similar strategies? Give some examples.

9. Consider the two expressions:

(a) 73 – 14 + 1 (b) 73 – 14 – 1

For each of these expressions, identify the expressions from the following collection that are equal to it.

(a) 73 – (14 + 1) (b) 73 – (14 – 1)

(c) 73 + (–14 + 1) (d) 73 + (–14 – 1)

Figure it Out *(Page 41 & 42)*

1. Fill in the blanks with numbers and boxes by signs, so that the expressions on both sides are equal.

(a) 3 × (6 + 7) = 3 × 6 + 3 × 7

(b) (8 + 3) × 4 = 8 × 4 + 3 × 4

(c) 3 × (5 + 8) = 3 × 5 $\square$ 3 × _____

(d) (9 + 2) × 4 = 9 × 4 $\square$ 2 × _____

(e) 3 × (______ + 4) = 3 ______ + ______

(f) (______ + 6) × 4 = 13 × 4 + ______

(g) 3 × (______ ______) = 3 × 5 + 3 × 2

(h) (______ + ______) × ______ = 2 × 4 + 3 × 4

(i) 5 × (9 – 2) = 5 × 9 – 5 × ______

(j) (5 – 2) × 7 = 5 × 7 – 2 × ______

(k) 5 × (8 – 3) = 5 × 8 ☐ 5 × ______

(l) (8 – 3) × 7 = 8 × 7 ☐ 3 × 7

(m) 5 × (12 – ______) = ______ ☐ 5 × ______

(n) (15 – ______) × 7 = ______ ☐ 6 × 7

(o) 5 × (______ – ______) = 5 × 9 – 5 × 4

(p) (______ – ______) × ______ = 17 × 7 – 9 × 7

2. In the boxes below, fill in '<', '>' or '=' after analysing the expressions on the LHS and RHS. Use reasoning and understanding of terms and brackets to figure this out, and not by evaluating the expressions.

(a) (8 – 3) × 29 ☐ (3 – 8) × 29

(b) 15 + 9 × 18 ☐ (15 + 9) × 18

(c) 23 × (17 – 9) ☐ 23 × 17 + 23 × 9

(d) (34 – 28) × 42 ☐ 34 × 42 – 28 × 42

3. Here is one way to make 14: 2 × (1 + 6) = 14. Are there other ways of getting 14? Fill them out below:

(a) ______ × (______ + ______) = 14

(b) ______ × (______ + ______) = 14

(c) ______ × (______ + ______) = 14

(d) ______ × (______ + ______) = 14

4. Find out the sum of the numbers given in each picture below in at least two different ways. Describe how you solved it through expressions.

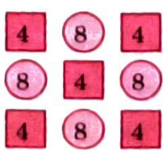


Figure it Out *(Page 42-44)*

1. Read the situations given below. Write appropriate expressions for each of them and find their values.

(a) The district market in Begur operates on all seven days of the week. Rahim supplies 9 kg of mangoes each day from his orchard, and Shyam supplies 11 kg of mangoes each day from his orchard to this market. Find the number of mangoes supplied by them in a week to the local district market.

(b) Binu earns ₹ 20,000 per month. She spends ₹ 5,000 on rent, ₹ 5,000 on food, and ₹ 2,000 on other expenses every month. What is the amount Binu will save by the end of a year?

(c) During the daytime, a snail climbs 3 cm up a post, and during the night, while asleep, accidentally slips down by 2 cm. The post is 10 cm high, and a delicious treat is on top. In how many days will the snail get a treat?

2. Melvin reads a two-page story every day except on Tuesdays and Saturdays. How many stories would he complete reading in 8 weeks? Which of the expressions below describes this scenario?

(a) 5 × 2 × 8
(b) (7 – 2) × 8
(c) 8 × 7
(d) 7 × 2 × 8
(e) 7 × 5 – 2
(f) (7 + 2) × 8
(g) 7 × 8 – 2 × 8
(h) (7 – 5) × 8

3. Find different ways of evaluating the following expressions:

(a) 1 – 2 + 3 – 4 + 5 – 6 + 7 – 8 + 9 – 10

(b) 1 – 1 + 1 – 1 + 1 – 1 + 1 – 1 + 1 – 1

4. Compare the following pairs of expressions using '<', '>', or '=,' or by reasoning.

(a) 49 – 7 + 8 ☐ 49 – 7 + 8

(b) 83 × 42 – 18 ☐ 83 × 40 – 18

(c) 145 – 17 × 8 ☐ 145 – 17 × 6

(d) 23 × 48 – 35 ☐ 23 × (48 – 35)

(e) (16 − 11) × 12 $\square$ −11 × 12 + 16 × 12

(f) (76 − 53) × 88 $\square$ 88 × (53 − 76)

(g) 25 × (42 + 16) $\square$ 25 × (43 + 15)

(h) 36 × (28 − 16) $\square$ 35 × (27 − 15)

5. Identify which of the following expressions are equal to the given expression without computation. You may rewrite the expressions using terms or removing brackets. There can be more than one expression which is equal to the given expression.

(a) 83 − 37 − 12
   (i) 84 − 38 − 12
   (ii) 84 − (37 + 12)
   (iii) 83 − 38 − 13
   (iv) −37 + 83 − 12

(b) 93 + 37 × 44 + 76
   (i) 37 + 93 × 44 + 76
   (ii) 93 + 37 × 76 + 44
   (iii) (93 + 37) × (44 + 76)
   (iv) 37 × 44 + 93 + 76

6. Choose a number and create ten different expressions having that value.

Exercise

Multiple Choice Questions

LEVEL - 1

Arithmetic Expressions

1. A class has 57 students and 19 desks. So, the number of students on each desk will be represented by expression
   (a) 57 − 19
   (b) 57 ÷ 19
   (c) 57 × 19
   (d) 57 + 19 *(NCERT Page 24)*

2. If ✡ = 2 and ☽ = 3 then which of the given expression represents

   2 3 3 2
   ✡ × ☽ − ☽ + ✡

   (a) 2 × 3 − 3 + 2
   (b) 2 × 3 − 2 + 3
   (c) 2 × 3 + 3 − 3
   (d) 3 × 2 + 2 + 3 *(Page 24)* An

3. Each pen has length 12 cm. Seven such pen placed end to end then length of 7 pens is represented as
   (a) 12 × 7
   (b) 12 ÷ 7
   (c) 12 + 12 × 7
   (d) 12 + 7 *(Page 24)* U

4. A book has 133 pages, Ravi reads 7 pages daily. Write an expression for the number of days needed to finish the book.
   (a) 133 − 7
   (b) 133 ÷ 7
   (c) 133 × 7
   (d) 133 + 7 *(Page 24)*

5. If 2 ✡ 2 = 4, 3 ✡ 3 = 9, 4 ✡ 4 = 16 then ✡ represents the operation
   (a) +
   (b) −
   (c) ×
   (d) ÷

Evaluating Arithmetic Expressions

6. Which operation is performed first to find the value of the expression 15 − 3 × 4 + 2?
   (a) Subtraction
   (b) Multiplication
   (c) Addition
   (d) Any of these *(Page 27)* U

7. Write an expression that represents the given situation "A plant grows 2 cm during day shrinks 1 cm at night. After 5 days how tall is it if it started at 2 cm".
   (a) 5 × (2 + 1)
   (b) (5 × 2 − 5 × 1)
   (c) 5 × 2 − 1
   (d) (5 × 2 − 5 × 1) + 2 *(Page 27)*

Quintal = 100kg.

8. Riya scored 15 marks in each of 4 tests and got a bonus 10 marks. Which of the given expression represents her total marks?
(a) 60 + 10
(b) 15 × 4 + 10
(c) 15 + 15 + 15 + 15 + 10
(d) All of these (Page 27) Ev

9. Write an expression representing subtraction of 5 times 8 from 7 times 10.
(a) 10 × 7 – 8 × 5
(b) 5 × 8 – 10 × 7
(c) 10 × 8 – 5 × 7
(d) 10 × 5 – 7 × 8 (Page 27) U

10. How many terms does the given expression have?

7 + 3 × 2 – 5

(a) 4 (b) 3
(c) 2 (d) 1 (Page 28) U

11. The expression 85 – 28 – 12 is equal to
(a) 80 – 23 – 7
(b) 85 – (28 – 12)
(c) 85 – (28 + 12)
(d) 84 – 29 – 12 (Page 30) U

12. Place brackets on correct place such that the given expression gives the value indicated.

65 – 2 × 18 – 12 = 53

(a) 65 – (2 × 18) – 12
(b) (65 – 2) × (18 – 12)
(c) 65 – 2 × (18 – 12)
(d) None of these (Page 30)

13. Which property is used in (3 + 4) + 5 = 3 + (4 + 5)?
(a) Commutative property
(b) Associative property
(c) Distributive property
(d) Both (a) and (b) (Page 31) An

14. Swapping the terms, represents the
(a) commutative property
(b) associative property
(c) addition property
(d) distributive property (Page 31) R

15. How many different expressions can be formed from a given value?
(a) Only one
(b) Only ten
(c) Depends on value
(d) Many (Page 32)

16. Which of the given expression will give the value 26?
(a) 17 + 2 × 3
(b) 15 + 12 – 1
(c) 30 – 17 × 2
(d) 13 × (5 – 2) (Page 35) Ev

17. Find the value of 45 – (20 + 5).
(a) 20 (b) 25
(c) 35 (d) 5 (Page 35) Ev

18. Which of the given expression shows the use of distributive property?
(a) 5 + (3 + 2) = (5 + 3) + 2
(b) 5 × (3 + 2) = 5 × 3 + 5 × 2
(c) 5 + (3 × 2) = 5 + 3 × 5 + 2
(d) 5 + 3 – 2 = 5 – 2 + 3 (Page 40) An

19. Which the following expression is equivalent to 98 × 12?
(a) 100 × 12 – 2 × 12
(b) 90 × 12 – 8 × 12
(c) 98 × 10 – 98 × 2
(d) None of these (Page 41) Ev

20. If ____ × (6 – 4) > 25 × (7 – 5) then number in the blank space will be
(a) 26 (b) 27
(c) 28 (d) All of these Ev

LEVEL - 2

21. Ramesh has subtracted 10 from a number and then add 3, what number he actually wants to subtract?
(a) 8 (b) 13 (c) 7 (d) 6 Ev

22. Terms in the expression 15 + 3 × 4 – 7 are:
(a) 15, 3, 4, –7 (b) 15, 12, –7
(c) 15, (3 × 4), –7 (d) 15, –3, 4, –7

23. Additive inverse of –15 is:
(a) –1 (b) 15 (c) 0 (d) –15
U

24. The value of expression 400 – (100 + 150) + (50 – 25) is
(a) 225 (b) 175 (c) 125 (d) 200
Ev

25. 95 × 28 is quickest evaluated using:
(a) 90 × 28 – 5 × 28
(b) 100 × 28 – 5 × 28
(c) 90 + 5 × 28
(d) 100 – 5 × 28

26. Which of the given expression has greatest value?
(a) 36 + 17 – 9 × 2 (b) 42 – 12 + 8
(c) 45 – 8 × 6 (d) 38 + 17 × 2 – 33

27. 1 – 2 + 3 – 4 + 5 – 6 + 7 – 8 + 9 – 10 is equal to
(a) –5 (b) 5
(c) –4 (d) 0
Ev

28. Find the value of 40 – (18 – 8) × 6 + 24.
(a) 204 (b) 260
(c) 44 (d) 4
Ev

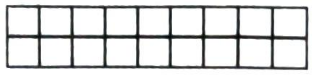
29. In a pattern of dots, first row has 3 dots, each new row has 2 more dots than previous one. Write an expression representing number of dots in 20th row.
(a) 20 + (3 × 2) (b) (20 × 3) + 2
(c) 3 + (20 × 2) (d) None of these

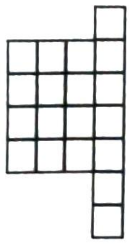
30. A teacher checks 20 papers per hours. In 3 hours, she check papers and leaves 15 unchecked. Write an expression for total number of papers.
(a) (20 × 3) + 15 (b) (20 × 3) – 15
(c) 20 + (15 × 3) (d) 20 – (15 × 3)
U

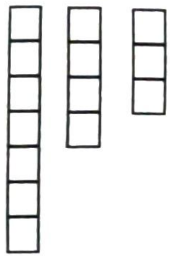
31. Which of the following applies both commutative and distributive property?
(a) (2 + 3) + 4 = 2 + (4 + 3)
(b) 4 × (2 + 6) = 4 × 6 + 4 × 2
(c) (2 + 4) × 5 = (4 + 2) × 5
(d) 4 × (3 + 2) = (4 × 3) + 2
An

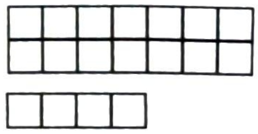
32. Place brackets on appropriate place so that the expression 16 – 2 × 5 – 3 + 4 represents the value 32.
(a) 16 – (2 × 5) – 3 + 4
(b) 16 – 2 × (5 – 3) + 4
(c) (16 – 2) × 5 – (3 + 4)
(d) (16 – 2) × (5 – 3) + 4

33. Which of the given arrangements matches with the expression 7 + (3 × 4)?

(a) 

(b) 

(c) 

(d) 

LEVEL - 3 (HOTS)

Competency Focused Questions (CFQs)

34. The value of 10 × 5 + (3 + 2) × (24 ÷ 6) is
(a) 74 (b) 62 (c) 70 (d) 84

35. Which of the following expressions has the value 63?
(a) 5 × (8 + 3) + 18 (b) 90 – (4 × 5 + 7)
(c) 100 – 3 × (9 + 3) (d) (6 + 7) × 5 + 8
Ev

36. If in the expression (25 ÷ 5) + (6 × 3) the first term of an expression is doubled and second term is halved, how does it affect the total sum?
(a) Increased by 3 (b) Decreased by 3
(c) Decreased by 4 (d) Increased by 4
Ap

37. Simplify : $48 \div 4 \times (5 + 7) - 6 \times 2 + (3 - 2) \times 5$
(a) 18 (b) 17
(c) 16 (d) 15 Ev

38. By placing brackets at different places how many different values the given expression will have?
$13 \times 5 - 6 + 2$
(a) 5 (b) 4 (c) 3 (d) 2

39. Which of the following expressions use all four operations and results in a prime number?
(a) $25 \div 5 + 4 \times 3 - 2$ (b) $16 \div 4 + 3 \times 2 - 1$
(c) $20 - 4 + (2 \times 5) \div 2$ (d) $30 \div 3 + 2 \times 4 - 5$ Ev

40. Which operation sign in the expression $12 + 8 \times 2 - 10$ should change to make the result maximum possible?
(a) + to × (b) × to ÷
(c) – to × (d) + to –

41. Which expression will not change the value if brackets are removed?
(a) $(8 + 2) \times 5$ (b) $20 - (5 + 3)$
(c) $18 \div (6 + 3)$ (d) $3 + (4 \times 5)$

42. In a classroom activity, there are 5 students and each student is assigned 3 sets of problems to solve. Each set contains 4 problems. The teacher decides to award 1 mark for each problem correctly solved. In addition, teacher gives 2 bonus marks for each set solved. 3 students complete the task whereas two students complete only 2 sets. If all solved questions was correct, then write an expression that represent total marks earned by all 5 students.
(a) $3 \times (3 \times 4 + 2) + 2 \times (2 \times 4 + 2)$
(b) $3 \times 4 \times 2 + 2 \times 4 \times 2$
(c) $3 \times (3 \times (4 + 2)) + 2 \times (2 \times (4 + 2))$
(d) $3 \times 4 \times 3 + 2 \times 2 \times 4$ U

Match the Following

In this section each question has two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (a), (b), (c) and (d) out of which, one is correct.

1. Match the expression given in List-I with their values given in List-II.

List-I	List-II
(P) $5 \times 6 + 2$	1. 15
(Q) $8 \times (3 + 2)$	2. 40
(R) $18 \div 3 + 9$	3. 13
(S) $12 - 5 + 6$	4. 32

(a) P-4, Q-2, R-3, S-1
(b) P-2, Q-4, R-3, S-1
(c) P-2, Q-4, R-1, S-3
(d) P-4, Q-2, R-1, S-3

2. Match the situation given in List I with the expression in List II.

List-I	List-II
(P) A taxi charges ₹ 20 as fixed charge and ₹ 12 per km, then expression representing charges for 12 km is	1. $(12 \times 12) + 20$
(Q) Expression representing area of rectangle, which is 12 cm more than area of square of side 20 cm.	2. $(20 \times 20) + 12$
(R) Riya saved ₹ 20 for 12 days and gave ₹ 12 to Megha after 12th day. Money that Riya had now can be expressed as	3. $12 \times (20 + 12)$
(S) A packet of chocolate cost ₹ 12 more than a packet of toffees costing ₹ 20. The price of 12 packets of chocolates can be represented as	4. $(20 \times 12) - 12$

(a) P-3, Q-2, R-4, S-1
(b) P-2, Q-3, R-1, S-4
(c) P-1, Q-2, R-4, S-3
(d) P-4, Q-1, R-3, S-2

Assertion & Reason Type

Directions : In each of the following questions, a statement of Assertion is given followed by a corresponding statement of Reason just below it. Of the statements, mark the correct answer as :

(a) If both assertion and reason are true and reason is the correct explanation of assertion.

(b) If both assertion and reason are true but reason is not the correct explanation of assertion.

(c) If assertion is true but reason is false.

(d) If assertion is false but reason is true.

1. **Assertion** : The expression $12 - (6 + 3)$ is same as $12 - 6 - 3$.

   Reason : While removing the bracket proceed by a negative sign, sign of terms inside the bracket changes.

2. **Assertion** : The value of the expression $32 - 6 \times 3 + 5$ is 208.

   Reason : In the expression $32 - 6 \times 3 + 5$, the terms are 32, $((-6) \times 3)$ and 5.

3. **Assertion** : The expression $(25 - 6) + 5$ is equal to $25 + (-6 + 5)$

   Reason : Grouping the terms of an expression in either of the following ways gives the same value

   (Term 1 + Term 2) + Term 3 = Term 1 + (Term 2 + Term 3)

4. **Assertion** : We can evaluate the value of an expression 25×45 by using the value of the expression 26×45, which is equal to 1170

   Reason : Multiplication takes precedence over addition in evaluating expression.

5. **Assertion** : Number of terms in the expression $25 - 4 \times 6 - 6 + 2 \times 3$ is 4.

   Reason : The term $6 + 2 \times 3$ is treated as single term.

Comprehension Type

PASSAGE-I : A water tank has 600 litres of water. Every day 30 litres of water is used by a person.

1. The expression representing the amount of water left in tank after 10 days is
   (a) $600 - (30 \times 10)$ (b) $(600 - 30) \times 10$
   (c) $(600 - 10) \times 30$ (d) $600 - (30 + 10)$

2. After how may days the tank will become empty?
   (a) 12 days (b) 15 days
   (c) 18 days (d) 20 days

PASSAGE-II : Additive inverse is also called the opposite of the number or negation of number or changed sign of original number. All subtractions in an expression should be rewritten as additions of inverses.

1. How should you rewrite $12 - 5 + 3 \times 4$ to identify its terms correctly?
   (a) $12 + (-5) + 3 + 4$
   (b) $12 + (-5) + (3 \times 4)$
   (c) $12 - 5 + (3 \times 4)$
   (d) $12 + (-5) + 12$

2. Which of the following are the terms of $7 + 2 \times 8 - 5$?
   (a) 7, 2, 8, –5 (b) 7, (2×8), –5
   (c) 7, $2 \times 8 - 5$ (d) 7, 16, –5

3. Which expression is equivalent to $17 - 6 + 3 \times 4 - 9$?
   (a) $17 + (-6) + (3 \times 4) + (-9)$
   (b) $17 - 6 + 3 \times 4 - 9$
   (c) $17 + 6 + 12 + 9$
   (d) $17 + (-6 + 3 \times 4 - 9)$

PASSAGE-III : A cinema hall sells 120 tickets for ₹ 150 each and 80 tickets for ₹ 100 each. Assume that all the tickets were sold.

1. Form an expression to represent the total money collected from ticket sales.
   (a) $(120 \times 80) + (150 \times 100)$
   (b) $(120 \times 150) + (80 \times 100)$
   (c) $(120 + 80) \times (150 \times 100)$
   (d) $(120 \times 100) + (150 \times 80)$

2. The amount collected from sales is
(a) ₹ 50000 (b) ₹ 24600
(c) ₹ 26000 (d) ₹ 24000

3. If the hall spends ₹5 per ticket sold in printing. What is the total expenditure on printing?
(a) ₹ 800 (b) ₹ 1000
(c) ₹ 1200 (d) ₹ 1500

Subjective Problems

Very Short Answer Type

1. Without evaluating the value of the expression, find which of the given expressions has greater value.
$265 + 352, 269 + 350$

2. Identify the number of terms in the given expression.
$12 \times (6 \div 2) - 3 + 6 \times 5$

3. Write a expression by using numbers 2, 5, 7, 9 and operators '+', '–', '×' and necessary brackets. Use one operator and number only once.

4. Check whether the values of the expressions $200 - (25 + 46)$ and $200 - 25 + 46$ are same?

5. "12 times 10 added to 15 times 5 and the result subtracted from 500." Write the arithmetic expression representing the given situation.

6. In a school assembly, there are boy and girls seated in rows. The boys are seated in 6 rows with 7 boys in each row. The girls are seated in 7 rows with 6 girls in row. Write an expression that represents the total number of students that are seated in assembly.

7. Evaluate the value of given expression
$2 - 3 + 4 - 6 + 6 - 9 + 8 - 12 + 10 - 15$

8. Find the correct placement of brackets for the expression $2 \times 6 - 5 - 3$ so that it gives the value 10.

9. Simplify the given expression by distributive property.
$6 \times (8 + 2)$

10. For a class activity a student needs
2 drawing sheets (₹ 5 each) 10 ₹
3 colours (₹ 10 each) 30 ₹
1 brush (₹ 12 each) 12 ₹
For a student, write an arithmetic expression to find the cost of supplies.

Short Answer Type

1. Rewrite the expression $25 - (6 + 2 - 7)$ by removing bracket and hence find its value.

2. Fill in the blanks.
(i) $24 - 16 = -16 +$ ....
(ii) $(2 - 6) + 5 = 2 + (-6 +$ ......$)$

3. Compare the following pairs of expressions using '<', '>' or '=' by reasoning.
(i) $(16 - 4) \times 10$ ______ $(12 - 3) \times 10$
(ii) $21 \times (7 + 3)$ ______ $(6 + 4) \times 21$

4. Verify the commutative property of addition for the following numbers.
(i) 15 and –6
(ii) –7 and –11

5. Verify the associative property of addition for the following numbers.
(i) –4, 6, and 7
(ii) –10, –2, and 15

6. Given: $53 \times 18 = 954$. Find the value of the following
(i) 55×18
(ii) 53×17

7. Rewrite the following expressions by removing brackets and then simplify them.
(i) $100 - (40 + 30 - 10)$
(ii) $55 + (25 - 20 + 35)$

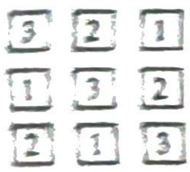
8. If the value of $2056 + 625 - 152$ is 2529 then without evaluating find the value of $2054 + 626 - 151$ by using given value.

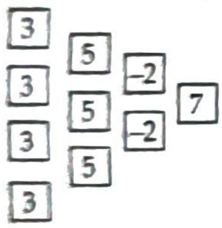
9. There are 12 groups of children in which there are 5 students in 6 groups and 3 students in 6 groups. Express the total number of students in form of arithmetic expression and hence find its value using the property.

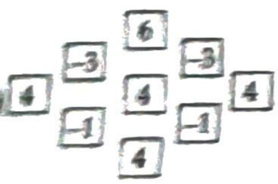
10. Find the value of the given expressions.
   (i) $11 + (6 - 3) \times 5$
   (ii) $42 \div 3 + 6 \times 5 - 4$

Long Answer Type

1. Add brackets at appropriate places in the expressions such that they lead to the value indicated.
   (i) $62 - 12 + 3 \times 4 = 2$
   (ii) $15 \times 3 + 2 - 5 = 70$
   (iii) $(-1) + 1 \times 2 + 5 = 0$
   (iv) $62 + 3 - 5 \times 3 + 3 = 53$
2. Fill in the blanks to get the indicated value.
   (i) ____ × (____ + ____) = 25
   (ii) (____ – ____) × (____ + ____) = 72
   (iii) (____ × ____) – ____ + ____ = 36
   (iv) (____ + ____) × ____ = 60
3. Observe the arrangements of number shown in the given pictures. Each shape contains some number. Using your observation skills, form two different algebraic expressions for each image to find the total sum of the numbers in each arrangement. Also find the sum of the numbers.

(i) 

(ii) 

(iii) 

4. Rekha runs a small food packaging business. She receives two different orders every day of the week.
   - In the first order, she packs 9 snack boxes each day from Monday to Friday, and 12 boxes each on Saturday and Sunday.
   - In the second order, she packs 11 meal boxes each day from Monday to Saturday but on Sunday, she packs only 5 meal boxes.

   Each box, whether snack or meal, takes 10 minutes to prepare.

   Using the given situation, answer the given questions.
   (i) Form an arithmetic expression to represent the total number of boxes Rekha packs in the entire week.
   (ii) Find the total time she spends preparing boxes during the week.
   (iii) How using arithmetic expressions helped you manage this complex situation efficiently?
5. If the value of $45 \times 112 = 5040$ then by using this value, find the values of given expressions.
   (i) 46×112
   (ii) 45×111
   (iii) $(54 - 9) \times (112 - 2)$
   (iv) 44×112

Integer/Numerical Value Type

1. If Renu buys 4 packets of pencils for ₹ 25 each and 2 erasers for ₹ 5 each and 1 sharpener for ₹ 10, then the amount she needs to pay is ₹______.
2. Change one term '9' in $(7 \times 3) + 9$ to double its value. Which term is being replaced by 9?
3. The value of the expression $9 \times (2 + 1) - 3$ is ______.
4. If the value of the expression $2 \times (6 + 3) - 5$ is p and value of $(5 \times 3) + (2 \times 4)$ is q, then $p + q$ is equal to ______.
5. If $54 \times 89 = 4806$. If 54 is increased by 2 and then multiply to 89 then the obtained value is ______.
6. Monica subtracts 100 from 500 and add 25 to the result obtained. Which number she actually wants to subtract?
7. If in the given expression $7 \times 4 + 3 \times 5$, 4 increased by 1 and 5 decreases by 1, then the new result will give the value ______.

8. A school organizes 3 buses with 42 seats each for a trip. If 99 students go for trip, then find how many seats were empty in each bus. (Assume there are equal empty seats in each bus).

9. The value of $100 - [60 + 5 + (18 - 6 \times 2)]$ is ______.

10. A group books 5 concert tickets at ₹1,200 each, pays ₹75 service charge per ticket and a one-time fee of ₹250. Total cost is ₹ ______.

Case Based Questions

Case I : A school canteen offers Sandwiches at ₹ 25 each, juice boxes for ₹ 15 each and fruit bowls at ₹ 30 each. One day a group of 4 students bought 2 sandwiches, 1 juice and 1 fruit bowl each, then one student returned a fruit bowl later and bought an extra juice box instead.

Based on the above information, answer the following questions. CFQs

1. Write an expression that represents the total bill for one student before the returning of a fruit bowl.
   (a) $2 + 25 + 15 \times 30$
   (b) $2 \times 25 + 15 + 30$
   (c) $(2 + 1 + 1) \times 25$
   (d) $25 + 15 + 30$

2. What is the total cost for all 4 students before any return?
   (a) ₹ 280 (b) ₹ 240 (c) ₹ 320 (d) ₹ 380

3. What is the cost difference due to the fruit bowl return and juice box purchase?
   (a) ₹ 15 increase (b) ₹ 15 decrease
   (c) ₹ 30 increase (d) No change

4. Which property justifies the given calculations $(2 \times 25) + (15 + 30) = (15 + 30) + (2 \times 25)$?
   (a) Commutative property
   (b) Associative property
   (c) Distributive property
   (d) None of these

5. Final bill for all 4 students after the change is
   (a) ₹ 310 (b) ₹ 280
   (c) ₹ 320 (d) None of these

Case II : King Omar gifted 120 gold coins each to his two sons - Prince Zayan and Rehan. Prince Zayan invested all his coins in a rare spice trade, and within a year, his wealth tripled. Prince Rehan spent half of his coins on a luxury ship and keep the rest with him. Within a year, the value of luxury ship investment doubled. However, a thief stole 20 coins from Rehan's hidden treasure.

Based on above information, answer the following questions. CFQs

1. The expression for Zayan's total wealth after the trade is
   (a) 3×120 coins
   (b) $3 + 120$ coins
   (c) $120 \div 3$ coins
   (d) $(120 \times 3) - 30$ coins

2. The expression for the number of coins left with Rehan after the theft is
   (a) $120 - (20 + 60)$ (b) $(120 \div 2) - 20$
   (c) $120 - 20$ (d) $120 \div (2 - 20)$

3. The expression for Rehan's total wealth after one year (ship value + remaining coins) is
   (a) $((120 \div 2) \times 2) + ((120 \div 2) - 20)$ coins
   (b) $120 \times 2 + 60 - 20$ coins
   (c) $(120 - 20) \times 2$ coins
   (d) $(120 \times 3) - 20$ coins

4. How many more wealth does Zayan have than Rehan in the end?

(a) 120 coins (b) 180 coins
(c) 200 coins (d) 220 coins

5. The combined total wealth of both princes after one year is

(a) 400 coins (b) 460 coins
(c) 560 coins (d) 520 coins

Case III : Riya, Diya and Tanya sell cupcakes at a school fest. Riya sells 6 trays with 4 cupcakes each at ₹ 15 per cupcake. Diya sells 3 trays with 5 cupcakes each at ₹ 10 per cupcake and Tanya sells 5 trays with 4 cupcakes each at ₹ 12 per cupcake.

Based on the above information, answer the following questions. CFQs

1. Write an expression that represent amount received by Tanya.

(a) ₹ 12 × (5 + 4) (b) ₹ (12 × 5 × 4)
(c) ₹ (12 × 5) + 4 (d) ₹ (12 × 4) + 5

2. Write an expression that represent number of cupcakes sold by Riya.

(a) 6 + 4 (b) 15 × 6
(c) 6 × 4 (d) 15 × 6 × 4

3. Write an expression that represent amount received by Diya from sales.

(a) ₹ 10 × (5 × 3) (b) ₹ 10 × (5 + 3)
(c) ₹ 5 × (10 + 3) (d) ₹ 3 × (5 + 10)

4. Who sold the maximum number of cupcakes?

(a) Riya (b) Diya
(c) Tanya (d) Both Riya and Diya

5. The total amount received by all three of them is

(a) ₹ 575 (b) ₹ 650
(c) ₹ 700 (d) ₹ 750